

2 Inheritance of flower colour and flower position in pea plants are controlled by two genes.

- Gene **P/p** controls flower colour. Allele **P** for purple flowers is dominant to allele **p** for white flowers.
- Gene **A/a** controls flower position. Allele **A** for flowers growing from the side of the shoot (axial position) is dominant to allele **a** for flowers growing at the end of the shoot (terminal position), as shown in Fig. 2.1.

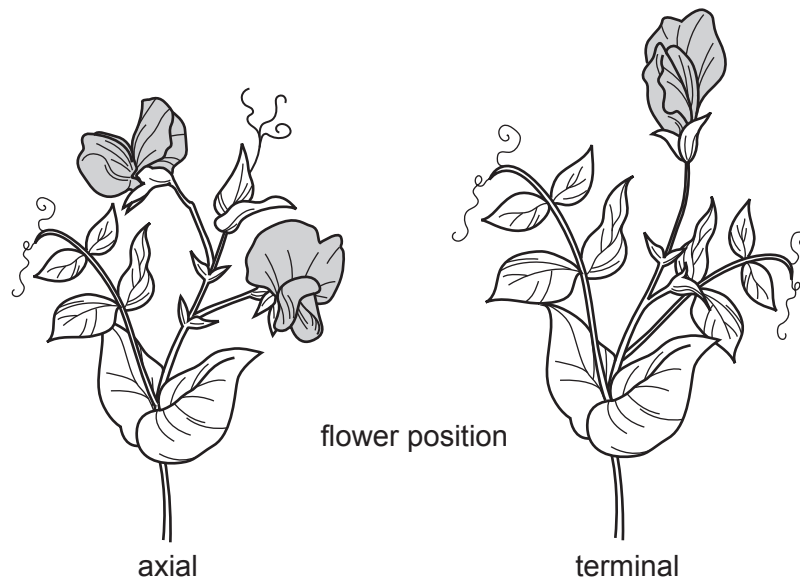


Fig. 2.1

A biologist predicted that, if the genes are on **different** chromosomes, the ratio of the phenotypes of the F₂ generation would be 9:3:3:1.

The biologist carried out a breeding experiment.

- Plants homozygous for white flowers and axial position were crossed with plants homozygous for purple flowers and axial position.
- All the F₁ plants had purple, axial flowers.
- The F₁ plants were crossed with each other.

Table 2.1 shows the results for the F₂ generation.

Table 2.1

F₂ phenotype	frequency
purple, axial flowers	1756
purple, terminal flowers	653
white, axial flowers	702
white, terminal flowers	234
total	3345

(a) The chi-squared test (χ^2 test) was used to analyse the data in Table 2.1.

(i) State **one** reason why the chi-squared test (χ^2 test) was used.

.....
 [1]

(ii) State the null hypothesis that the biologist would use for this test.

.....

 [1]

(iii) Complete Table 2.2 **and** calculate the value of χ^2 for the results of the F₂ generation.

The equation for the calculation of χ^2 is:

$$\chi^2 = \sum \frac{(O - E)^2}{E}$$

O = observed result

E = expected result

$\sum$ = sum of

Table 2.2

offspring phenotype	<i>O</i>	<i>E</i>	$\frac{(O - E)^2}{E}$
purple, axial flowers	1756		
purple, terminal flowers	653		
white, axial flowers	702		
white, terminal flowers	234		
$\chi^2 =$			

[3]

Table 2.3 shows some critical values of χ^2 at different probability levels.

Table 2.3

degrees of freedom	probability (p)			
	0.10	0.05	0.01	0.001
1	2.71	3.84	6.64	10.83
2	4.61	5.99	9.21	13.82
3	6.25	7.82	11.34	16.27
4	7.78	9.49	13.28	18.46

(iv) State the critical value at $p < 0.05$ for this χ^2 test. [1]

(v) Use your calculated value of chi-squared (χ^2) to:

- explain whether the null hypothesis should be accepted or rejected

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- suggest a conclusion the biologist could make about the inheritance of the genes controlling flower colour and flower position in pea plants.

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 [3]

[Total: 9]